

# O9V - FASTWIND line profiles

$R$  = intrinsic |  $v \sin i = 0 \text{ km s}^{-1}$  |  $S/N$  = none |  $v_{\text{turb, formal}} = 10 \text{ km s}^{-1}$

| Stellar & wind parameters             |                                                          |                                                 |
|---------------------------------------|----------------------------------------------------------|-------------------------------------------------|
| <b>Stellar</b>                        | <b>Wind</b>                                              | <b>Atmosphere</b>                               |
| $T_{\text{eff}} = 32,882 \text{ K}$   | $\dot{M} = 6.6 \times 10^{-8} M_{\odot} \text{ yr}^{-1}$ | $v_{\text{turb, model}} = 10 \text{ km s}^{-1}$ |
| $\log g = 3.92$                       | $v_{\infty} = 2300 \text{ km s}^{-1}$                    | $\xi_{\text{He}} = 2$                           |
| $R_{*} = 7.53 R_{\odot}$              | $\beta = 0.8$                                            | $f_{\text{cl, max}} = 10$                       |
| $R_{\text{max}} = 120 R_{*}$          | $v_{\text{min}} = 0.1 \text{ km s}^{-1}$                 | $v_{\text{cl, start}}/v_{\infty} = 0.05$        |
| $T_{\text{min}}/T_{\text{eff}} = 0.6$ | $v_{\text{trans}} = 0.1 a_s$                             | $v_{\text{cl, max}}/v_{\infty} = 0.1$           |
| <b>Derived parameters</b>             |                                                          |                                                 |
| $\log(L/L_{\odot}) = 4.776$           | $M_{\text{spec}} = 17.20 M_{\odot}$                      | $\log Q = -13.536$                              |

| Abundances             |                              |
|------------------------|------------------------------|
| <b>Global</b>          | <b>Individual</b>            |
| $Z = 1 Z_{\odot}$      | $\epsilon_{\text{C}} = 8.33$ |
| $N_{\text{He}} = 0.11$ | $\epsilon_{\text{N}} = 8.46$ |
|                        | $\epsilon_{\text{O}} = 8.59$ |

